Lab 2: REST API and Asynchronous File Queue

Purpose and architecture

This project provides four REST operations: a greeting, UTC time, job submission, and job status lookup. REST uses HTTP methods, resource URLs, JSON representations, and meaningful response codes. The API accepts work without waiting for it to finish.

Queue and asynchronous execution

POST /api/tasks validates the input, saves a task record, writes a JSON message to the queue, and returns HTTP 202 with a UUID. The separate worker claims the message, marks the task processing, converts its text to uppercase, saves the completed result, and only then deletes the claimed message. GET /api/tasks/{id} returns the current status and result.

Worker interruption and recovery

When the worker is stopped, new tasks remain queued. When it starts again, it resumes processing. Messages claimed by a previous crashed worker are returned to the queue on startup. Since uppercase conversion is idempotent, replaying a task is safe for this demonstration. A production deployment would require stronger concurrency and durability guarantees.

Validation and expected responses

GET /api/message and GET /api/time return 200. POST /api/tasks with valid text returns 202. Invalid JSON or text returns 400. GET /api/tasks/unknown returns 404. A task changes from queued to processing and completed. The implementation includes automated tests and a latency-check script.

Verification evidence — to be captured on the student computer

Screenshots of live HTTP responses are intentionally not fabricated. Start app.py and worker.py, perform the commands in README.md, and save screenshots for: message 200, time 200, invalid task 400, unknown task 404, task accepted 202, queued status with worker stopped, and completed status after worker restart. Place them in screenshots/.

Limitations

This is a local teaching implementation using a filesystem queue. It has no authentication, distributed queue, or database. Performance targets are specified in spec.md and must be measured on the actual machine before claiming they have been met.